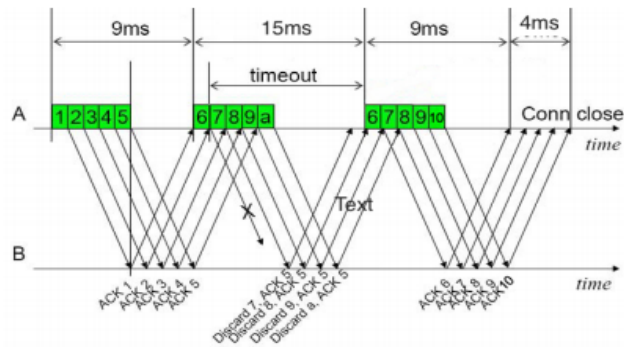
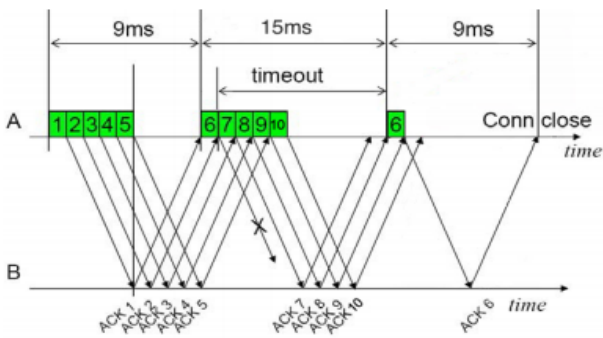


(a) (15 Points) GBN with packet 6 lost.

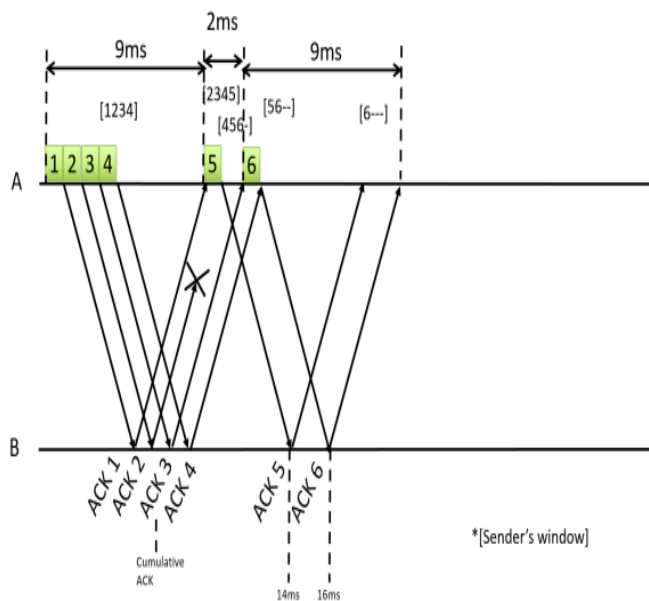


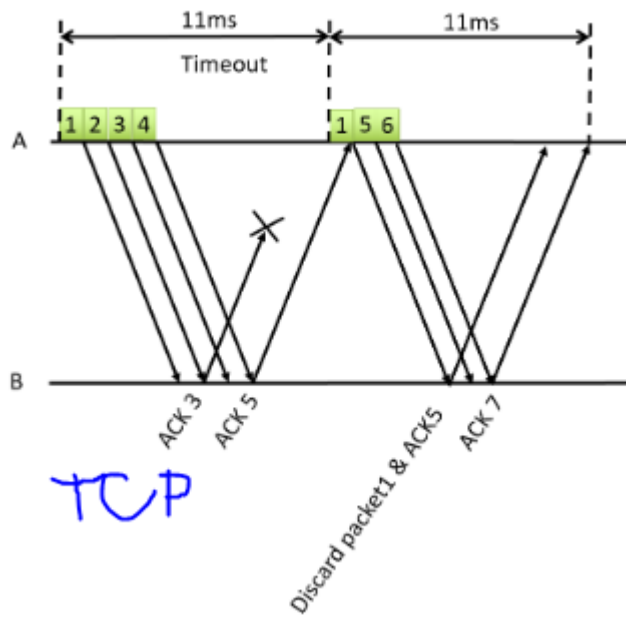
(b) (15 Points) SR with packet 6 lost.



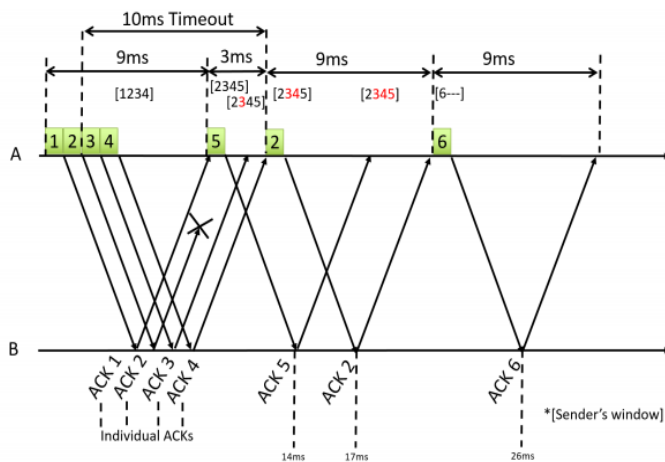
(a) (15 Points) GBN with ACK 2 lost.

Total time = 20 ms

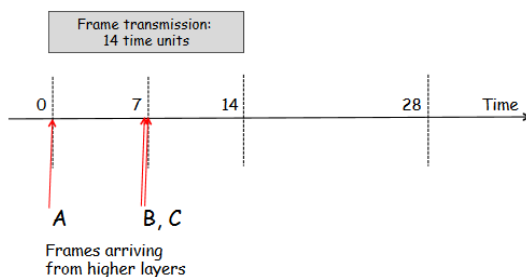




(b) (15 Points) Selective Repeat with ACK 2 lost.
Total time = 30 ms

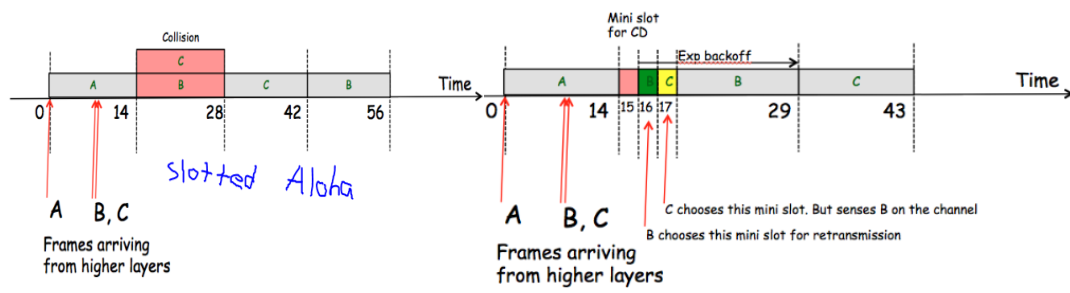


2. (20 Points) Random Access Protocols. Consider hosts A,B,C that transmit on a shared medium, using a random access protocol. Each host has exactly one frame to send, with transmission time = 14 time units. Host A wants to transmit at time $t = 0$. Hosts B,C both want to transmit at time $t = 7$, as shown in the figure below. Time is slotted and all hosts are synchronized to the clock.

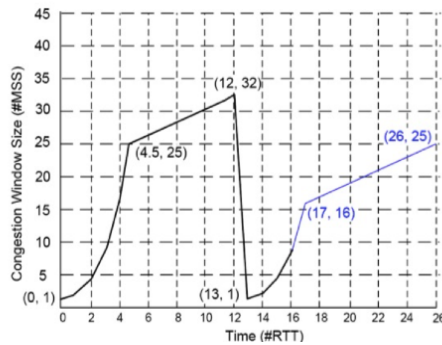


If needed to make randomized decisions, each host has access to the following sequence of random numbers, drawn uniformly at random between 0 and 1:

- Station A draws numbers: 0.45, 0.11, 0.71....
- Station B draws numbers: 0.10, 0.65, 0.91....
- Station C draws numbers: 0.83, 0.41, 0.25



- (a) (5) Identify the intervals of time when TCP slow start is operating. What is the respective SSThres (slow start threshold) in TCP congestion control?
 Answer: (0,4.5) and (13, 16). The SSThres are 25 and 16, respectively.
- (b) (2 Points) Identify the interval(s) of time when TCP congestion avoidance is operating. What is the respective SSThres?
 Answer: (4.5, 12). The SSThres is 25.
- (c) (3 Points) How many segments are lost from time 0 to 16?
 Answer: 1
- (d) (10 Points) Assume that no packet is lost during time 16-26. Complete the TCP congestion window curve from time 16 to 26.
 Answer: The critical information in the curve was the y-coordinate of the turning point in what you add, which is the indicator of the SSThres.



Slotted ALOHA: analysis

efficiency: long-run fraction of successful slots (assume many nodes, all with many frames to send)

- ❖ suppose: N nodes with many frames to send, each transmits in slot with probability p
- ❖ prob that a given node has success in a slot = $p(1-p)^{N-1}$
- ❖ Efficiency $E(p,N)$ = prob that any node has a success = $Np(1-p)^{N-1}$
- ❖ max efficiency: $p^* = 1/N$ maximizes $Np(1-p)^{N-1}$
 - take derivative $dE/dp=0$
- ❖ for many nodes, take limit of $Np^*(1-p^*)^{N-1}$ as N goes to infinity, gives:
max efficiency = $1/e \approx .36$

at best: channel used for useful transmissions 36% of time!

Another 36% are empty and ~28% go to collisions

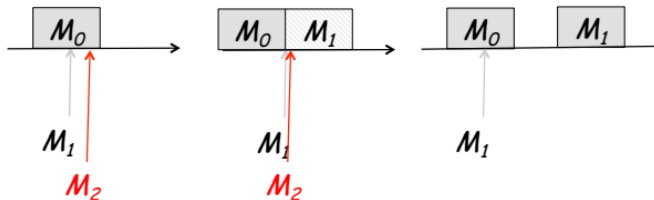
Link Layer 5-34

CSMA (carrier sense multiple access)

CSMA: listen before transmit:

if channel sensed idle: transmit entire frame

- ❖ if channel sensed busy, defer transmission
- ❖ human analogy: **dont interrupt others!**

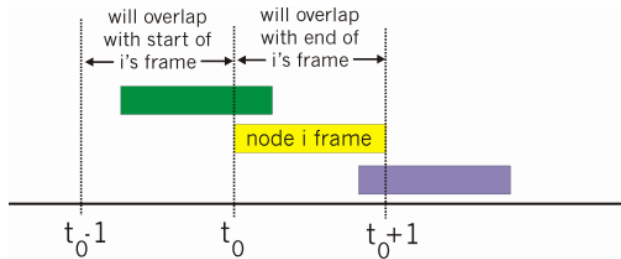


- ❖ One would hope that CS eliminate collisions. It doesn't!
 - Because of protocol (**other users synchronization**)
 - Because of physics (sensing is not instantaneous)

Link Layer 5-34

Pure (unslotted) ALOHA

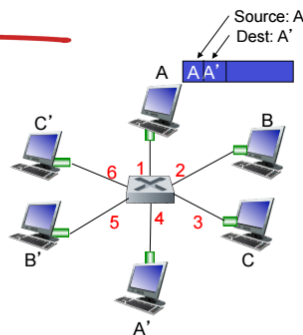
- ❖ unslotted Aloha: simpler, no synchronization
- ❖ when frame first arrives
 - transmit immediately
- ❖ If the message collides, try sending "later"
 - Wait a random time $\sim$ transmit in the next time frame w.p. p
- ❖ **Q:** is collision probability higher than in slotted Aloha?
 - A: frame sent at t_0 collides with other frames sent in $[t_0-1, t_0+1]$



Link

Switch: self-learning

- ❖ **switch learns** which hosts can be reached through which interfaces
 - when frame received, **switch** "learns" location of sender: incoming LAN segment
 - records sender/location pair in **switch** table

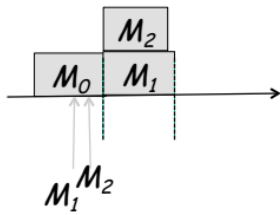


MAC addr	interface	TTL
A	1	60

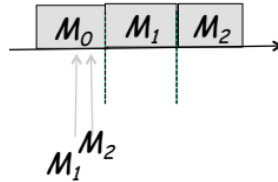
Switch table
(initially empty)

CSMA: Carrier Sensing and Persistence

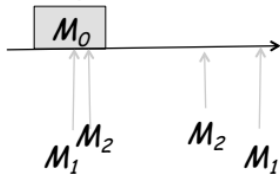
(1-) **Persistent**: defer transmission until channel becomes idle



P-Persistent: once the channel becomes idle, transmit with prob. p ; or defer until next slot



(0-) **Non-Persistent**: defer transmission, sense again after a random time

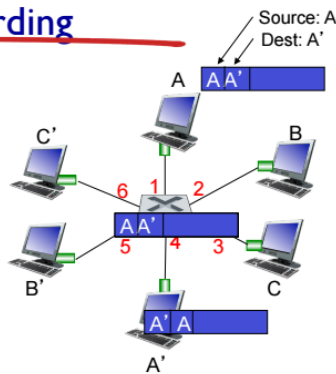


Choice of p : throughput vs delay

Data Link Layer

Switch: filtering, forwarding

- ❖ frame destination, A' , location unknown: **flood**
- ❖ destination A location known: **selectively send on just one link**



MAC addr	interface	TTL
A	1	60
A'	4	60

switch table
(initially empty)